AUTOMOTIVE DEALERSHIP ERP & BUSINESS PLATFORM

Functional Requirements & Project Planning Document

A comprehensive planning guide for a vehicle dealership and import company that sells, imports, brokers, and swaps cars and other vehicle categories.

Platform scope	Public website, customer portal, internal ERP, automotive services
Business model	Vehicle sales, imports, brokerage, trade-ins, swaps, and customer services
Document status	Initial functional scope for planning and development

Prepared as a structured reference for product planning, requirements gathering, and ERP design.

Document Overview

This document organizes the requested automotive website menu and expands it into functional requirements for the public website, customer portal, and internal ERP. It is a business and functional planning document—not a codebase or final technical architecture.

- Public website and navigation structure
- Vehicle product categories and inventory requirements
- Automotive services and operational workflows
- Company content, careers, and contact management
- ERP modules and system integrations
- Business rules, roles, and implementation recommendations

1. Project Vision & Platform Structure

The organization needs a connected automotive business platform that supports vehicle sales, vehicle importation, brokerage, trade-ins, swaps, customer enquiries, and internal business operations.

1.1 Main platform areas

Platform area	Purpose
Public-facing website	Promote the company, display approved inventory, explain services, publish company content, and generate customer leads.
Customer portal	Allow customers to submit enquiries, save comparisons, request sourcing, track eligible requests, and manage their profile.
Internal ERP	Manage inventory, sales, imports, customers, finance, employees, approvals, and reporting.
Automotive services	Manage AI matching, loan estimates, brokerage, customs services, vehicle sourcing, and sell-or-swap requests.

2. Public Website Navigation

Recommended navigation based on the requested menu items:

Menu	Items
Products	Executive Class; Sport & Performance; Premium SUVs; Electric Vehicles; Power Bikes; Executive Buses
Services	AI Car Match; Loan Calculator; Compare Cars; Auto Brokers; Customs Specialists; Sell or Swap
Company	News & Events; About Us; Careers; Contact
Recommended supporting navigation	Home; Inventory / Browse All Cars; Book a Test Drive; My Account; Sign In / Register

Navigation recommendation: Keep the main menu concise. Use Products, Services, and Company as dropdowns or mega-menu groups. Place “Sell or Swap” as a prominent call-to-action if it is a major revenue-generating service.

3. Products Module

The Products section should be connected directly to ERP inventory. Public listings should use approved vehicle records from the inventory database rather than a separate manually maintained catalogue.

3.1 Vehicle categories

Category	Scope / examples
Executive Class	Luxury and executive vehicles for business executives and premium buyers; e.g. Mercedes-Benz S-Class, BMW 7 Series, Lexus LS.
Sport & Performance	Sports cars, performance vehicles, supercars, and performance-oriented automobiles.
Premium SUVs	Luxury SUVs, family SUVs, off-road vehicles, and premium utility vehicles.
Electric Vehicles	Fully electric vehicles and electric SUVs; optionally include hybrid vehicles as a separate subcategory.
Power Bikes	Motorcycles, sports bikes, touring bikes, superbikes, and other power bikes.
Executive Buses	Luxury buses, executive coaches, corporate transport vehicles, and commercial passenger buses.

3.2 Product management functions

- Create, edit, archive, and remove vehicle listings with appropriate audit controls.
- Assign vehicles to categories and subcategories.
- Upload multiple images, videos, specifications, and supporting documents.
- Define make, model, year, trim, condition, and category-specific specifications.
- Set prices in USD, NGN, or configured currencies; manage exchange-rate assumptions where needed.
- Manage vehicle availability, reservations, featured listings, and promotions.
- Track vehicles in transit, awaiting clearance, in inspection, or in stock.
- Publish or unpublish listings from the public website.
- Manage listing descriptions, SEO metadata, and customer-facing specifications.

3.3 Core vehicle data fields

Field	Description
Stock ID	Unique internal inventory identifier.
VIN / chassis number	Vehicle identification; apply access controls to sensitive identifiers.
Category	Executive, SUV, EV, power bike, bus, etc.
Make / model / trim	Manufacturer and vehicle model information.
Model year	Model or manufacturing year, clearly labelled.
Mileage	Distance and unit (kilometres or miles).
Condition	Brand-new, foreign-used, locally-used, or other company-defined condition.
Transmission / fuel	Transmission type, petrol, diesel, electric, hybrid, etc.
Colours	Exterior and interior colour details.
Purchase cost	Acquisition price and currency.
Import / landed cost	Shipping, duties, clearance, port, and related costs.
Selling price	Approved customer-facing price.
Margin	Estimated and realized gross margin.
Location	Branch, showroom, warehouse, port, or transit location.

Field	Description
Status	Draft, in transit, clearance, inspection, available, reserved, sold, or archived.

4. Services Module

The Services area should create new business opportunities beyond direct vehicle sales. Each service should have a public information page, enquiry or request form, internal workflow, assignment process, and reporting.

4.1 AI Car Match

Purpose: help customers discover vehicles that fit their preferences and budget.

- Collect budget, preferred category, brands, vehicle use, passenger capacity, fuel type, new/used preference, country of origin, financing interest, and comfort/performance/safety priorities.
- Match customer preferences against current inventory and approved vehicle availability.
- Display recommended vehicles, price, specifications, and a concise explanation of the match.
- Allow customers to compare recommendations, request a quote, or contact sales.
- Capture matching requests as CRM leads and retain consent and preference data appropriately.

Implementation approach: Start with transparent rules-based matching and consider AI-powered recommendations later. The recommendation engine must use real inventory, current pricing, and availability.

4.2 Loan Calculator

Purpose: provide an indicative vehicle financing estimate and capture financing enquiries.

- Inputs: vehicle price, deposit, principal, interest rate, loan term, currency, and optional fees.
- Outputs: down payment, amount financed, estimated monthly payment, total interest, total repayment, and estimated balance.
- Allow the customer to submit the estimate as a financing enquiry.
- Store the enquiry and assign it to an authorized sales or finance employee.
- Maintain financing partner details and record actual financing terms separately from calculator estimates.

For a standard fixed-rate monthly repayment, the formula is:

$$M = P \times [r(1 + r)^n] / [(1 + r)^n - 1]$$

Where M is monthly repayment, P is principal, r is monthly interest rate, and n is the number of monthly payments. If the interest rate is zero, monthly payment is principal divided by number of payments. The calculator must label results as estimates, not approved loan offers.

4.3 Compare Cars

- Compare multiple vehicles side by side.
- Compare price, make/model, year, mileage, engine, fuel type, transmission, horsepower, safety, features, warranty, condition, and estimated financing.
- Add or remove vehicles and highlight differences.
- Share comparison results or request a quotation.
- Only compare vehicles with sufficient data; clearly mark unavailable specifications.

4.4 Auto Brokers

Purpose: manage customers who need the company to source vehicles on their behalf.

- Register sourcing requests with vehicle requirements, budget, preferred market, and target timeframe.

- Assign a broker and track supplier, market, or auction research.
- Present options and quotations to the customer.
- Record customer approval, purchase authorization, brokerage fees, and payment milestones.
- Track acquisition, inspection, shipping, delivery, and completion.
- Manage broker profiles, commissions, performance, and customer communication history.

Broker workflow stage	Expected activity
Request received	Customer submits sourcing requirements; request enters CRM.
Broker assigned	Authorized manager assigns a broker.
Sourcing	Broker searches markets, suppliers, or approved auction sources.
Options presented	Vehicle options and estimates are shared with customer.
Approval and agreement	Customer approves a vehicle and quotation; terms are documented.
Purchase and logistics	Purchase authorization, acquisition, inspection, and shipping are tracked.
Delivery and closure	Vehicle is delivered; costs, fees, and customer satisfaction are recorded.

4.5 Customs Specialists

Purpose: coordinate vehicle importation, shipping, customs documentation, clearance, and delivery.

- Register import requests and customer/company ownership.
- Record vehicle origin, supplier, purchase invoice, and shipment information.
- Track vessel, container, shipping agent, and expected arrival where applicable.
- Store invoices, shipping documents, customs paperwork, and other import records.
- Record customs declaration references and estimated/actual duties and taxes.
- Track port handling, clearance status, inspection, and delivery after clearance.
- Reconcile estimated costs against actual shipping, customs, and port charges.
- Maintain an audit trail for expenses, documents, and clearance updates.

Customs, tax, and import rules vary by jurisdiction and vehicle type. The ERP should support configurable rules and professional review rather than hardcoding duty estimates as guaranteed amounts.

4.6 Sell or Swap

Purpose: allow customers to sell a vehicle to the company or exchange it for another vehicle.

Sell a vehicle — customer flow

- Customer submits vehicle details, images, VIN/chassis information, mileage, condition, expected price, and ownership documentation.
- Company schedules inspection and performs valuation.
- Company issues a preliminary or final offer with validity period and terms.
- Customer accepts or rejects the offer.
- After acceptance, manage documentation, payment, ownership transfer, and inventory intake.

Swap a vehicle — customer flow

- Customer selects a vehicle from available inventory and submits their current vehicle details.
- Company inspects and values the trade-in vehicle.

- ERP calculates the difference between target vehicle price, trade-in value, and approved fees or adjustments.
- Customer receives a swap quotation and accepts or rejects it.
- After approval and payment/refund settlement, complete documentation, ownership transfer, and inventory updates.

Illustrative swap calculation	Amount
Target vehicle selling price	■25,000,000
Customer trade-in valuation	■12,000,000
Additional approved fees	■500,000
Customer balance payable	■13,500,000

This is an example only. The system should support discounts, valuation adjustments, additional charges, approval limits, and cases where the company owes the customer a balance.

- Trade-in request and inspection management.
- Valuation history and approval workflow.
- Swap quotations and customer acceptance.
- Customer balance, payment, refund, and receipt tracking.
- Ownership transfer checklist and document storage.
- Inventory intake for acquired trade-in vehicles.
- Swap profitability and transaction reporting.

5. Company Module

5.1 News & Events

Purpose: publish company news, vehicle arrivals, industry articles, launches, exhibitions, campaigns, and announcements.

- Create, edit, schedule, publish, unpublish, and archive content.
- Upload featured images, galleries, and videos.
- Categorize news and events and manage SEO metadata.
- Feature important announcements on the homepage.
- Optionally manage event registration and attendee enquiries.

5.2 About Us

The About Us page should communicate company identity and credibility.

- Company overview, mission, vision, values, history, and experience.
- Dealership and import expertise, brokerage services, and customer commitments.
- Business partnerships, certifications, and relevant affiliations.
- Showroom and branch locations.
- Content editable through the CMS by authorized staff.

5.3 Careers

Purpose: publish vacancies and manage recruitment.

- Publish vacancies with department, location, employment type, responsibilities, and requirements.
- Allow candidates to search and filter jobs.

- Accept applications, CVs, and supporting documents.
- Send application confirmation and track candidate status.
- Manage shortlisting, interviews, feedback, hiring decisions, and recruitment history.

Potential departments include Sales & Marketing, Importation, Customs & Logistics, Finance, Customer Service, IT, Administration, and Human Resources.

5.4 Contact

- Display company phone numbers, business email, physical and branch addresses, business hours, map links, and social media.
- Provide WhatsApp contact where appropriate.
- Collect full name, email, phone, subject, enquiry category, message, and optional vehicle reference.
- Store enquiries in CRM and assign them to the correct department or employee.
- Track status, assigned employee, response history, and follow-up date.

Suggested enquiry categories: vehicle purchase, import, sell or swap, financing, brokerage, customs clearance, careers, and general enquiry.

6. Website-to-ERP Integration Map

Public website feature	ERP module	Business purpose
Products and vehicle categories	Inventory management	Manage stock, listings, and vehicle availability.
AI Car Match	AI/customer matching + CRM	Recommend available vehicles and generate leads.
Loan Calculator	Finance + CRM	Generate financing enquiries; record estimates.
Compare Cars	Inventory + website	Compare vehicle records and specifications.
Auto Brokers	Brokerage management + CRM	Manage sourcing requests, fees, and broker assignments.
Customs Specialists	Import and logistics	Track import requests, documents, clearance, and expenses.
Sell or Swap	Trade-in + inventory + finance	Manage valuation, exchange, settlement, and stock intake.
News & Events	CMS	Manage public content and event information.
About Us	CMS	Manage corporate information.
Careers	HR/recruitment	Manage vacancies and candidate applications.
Contact	CRM	Capture, assign, and track enquiries.

7. Core Internal ERP Modules

Module	Key responsibilities
Dashboard & analytics	Sales, revenue, stock, profitability, imports, leads, and operational KPIs.
Inventory & vehicle management	Vehicle records, stock movements, branches, reservations, valuation, and availability.
Sales & order management	Quotes, orders, discounts, approvals, customer payments, receipts, delivery, and sales records.

Module	Key responsibilities
Import & logistics	Supplier purchases, shipping, customs, port expenses, landed costs, and import tracking.
Trade-in & swap	Vehicle valuation, inspection, approval, swap quotation, settlement, and acquired stock.
CRM & customer management	Customer profiles, leads, enquiries, follow-ups, broker/customer communication, and history.
Accounting & finance	Income, expenses, invoices, receipts, accounts, taxes, financial reconciliation, and reports.
HR & employee management	Employee profiles, recruitment, attendance, payroll integration, and staff records.
Roles & permissions	Access control by role, branch, department, and approval authority.
Document management	Vehicle documents, import records, invoices, contracts, customer files, and secure document access.
CMS	Website pages, news, events, careers, SEO, and publication workflows.

8. Recommended Business Workflows

8.1 Lead-to-sale

Step	Process
1	Customer browses inventory, uses AI Car Match, or submits an enquiry.
2	CRM creates a lead and assigns it to sales.
3	Sales staff contact customer, qualify requirements, and prepare quotation.
4	Vehicle availability and price are confirmed; reservation rules are applied.
5	Customer approves order and completes required payment.
6	ERP records sale, updates inventory, generates documents, and schedules delivery.
7	Accounting records transaction and management reports update.

8.2 Import-to-stock

Step	Process
1	Create import purchase/request and supplier record.
2	Record vehicle details, purchase price, currency, and authorization.
3	Track shipping, insurance, vessel/container, and expected arrival.
4	Record customs documentation, duty, tax, port, and clearance costs.
5	Complete inspection, compliance checks, and internal receiving.
6	Calculate actual landed cost and update stock location/status.
7	Approve vehicle listing for public inventory.

8.3 Sell-or-swap workflow

Step	Process
1	Customer submits sell or swap request.

Step	Process
2	Staff review vehicle details and schedule inspection.
3	Valuation is prepared and approved according to authority limits.
4	Company issues offer or swap quotation.
5	Customer accepts; documents and payment/refund terms are confirmed.
6	Ownership transfer and vehicle handover are completed.
7	Acquired vehicle enters inventory; accounting records settlement.

9. Business Rules & Controls

- A vehicle must not be sold or reserved by multiple customers at the same time.
- Reservations should have an expiry date, deposit rules, and cancellation policy.
- Each vehicle must have a unique internal stock identifier.
- Purchase cost, shipping, customs, port charges, repairs, and other costs should be separately traceable.
- Vehicle profitability should use actual costs where available, not only purchase price.
- Trade-in vehicles must undergo inspection and valuation approval.
- Financing calculator results must not be represented as guaranteed loan approvals.
- Sales discounts, refunds, and financial adjustments should require approval based on configured limits.
- Only approved inventory should be published on the public website.
- Important financial, stock, pricing, and document changes must be logged.
- Import costs should be reconciled against actual invoices and charges.
- Staff should access only the ERP modules and records authorized for their roles.
- Customer and vehicle documents should have appropriate privacy and access controls.

10. User Roles & Access

Role	Typical access
Super Administrator	System configuration, users, permissions, and all authorized business areas.
Management	Company dashboards, approvals, operational reports, and oversight.
Sales Manager	Sales team, quotations, sales approvals, leads, and vehicle availability.
Sales Representative	Assigned leads, customer enquiries, quotations, and permitted inventory information.
Inventory Officer	Vehicle records, stock movements, receiving, inspection, and reservations.
Import/Logistics Officer	Import requests, shipping, customs documents, clearance, and import costs.
Broker	Assigned sourcing requests, supplier options, quotations, and broker workflow.
Accountant/Finance Officer	Invoices, payments, expenses, reconciliation, and finance reports.
HR Officer	Recruitment, candidates, employee records, and HR workflows.
Content Manager	Website pages, news, events, careers content, and publication.
Customer	Own profile, enquiries, saved comparisons, and eligible request tracking.

11. Reporting & Analytics

- Sales by period, branch, salesperson, category, make, and model.
- Inventory availability, ageing, reservations, and stock valuation.
- Vehicle purchase cost, landed cost, selling price, and gross margin.
- Import status, shipping delays, customs expenses, and clearance time.
- Broker sourcing requests, completed transactions, fees, and commissions.
- Trade-in and swap valuations, settlement, and profitability.
- Customer leads, enquiry sources, response times, and conversion rates.
- Loan calculator submissions and financing enquiries.
- Website product views, enquiries, and conversion performance.
- Expenses, revenue, receivables, payables, and cash-flow reporting.

12. Implementation Recommendations

Build the core operational modules first and connect public features to reliable ERP data. This reduces duplicated records and makes the platform easier to maintain.

Phase	Recommended scope
Phase 1 — Core foundation	User authentication, roles, company settings, vehicle categories, inventory, branches/locations, CRM, and document management.
Phase 2 — Dealership operations	Sales, quotations, reservations, payments, accounting foundation, import/logistics, and approval workflows.
Phase 3 — Customer-facing services	Public inventory website, contact forms, compare cars, loan calculator, and customer portal.
Phase 4 — Specialized automotive services	Sell or swap, trade-in inspection and valuation, auto brokers, and customs specialists.
Phase 5 — Intelligence and growth	AI Car Match, advanced reporting, marketing analytics, and automation.
Phase 6 — Optimization	Security review, performance testing, user acceptance testing, operational training, and iterative improvements.

12.1 Key design considerations

- Use one source of truth for vehicle inventory and pricing.
- Keep public website access separate from internal ERP permissions.
- Design category-specific vehicle specifications while supporting common vehicle fields.
- Support multiple branches, warehouses, currencies, and vehicle locations if the company requires them.
- Use configurable approval limits for sales, expenses, discounts, valuations, and refunds.
- Make workflow statuses and required documents configurable where business processes may change.
- Keep a full audit history for important business transactions.
- Plan for secure document storage, backups, access controls, and recovery procedures.
- Use realistic test data and perform end-to-end testing before production deployment.

13. Final Project Scope Summary

Business area	Main responsibility
Automotive website	Vehicle marketing, product discovery, company information, and lead generation.
Dealership ERP	Internal business operations, inventory, sales, finance, staff, and reporting.
Automotive services	Brokerage, customs coordination, vehicle sourcing, financing assistance, and trade-ins.
Customer portal	Customer profiles, enquiries, comparisons, and eligible transaction/request tracking.

Recommended foundation: inventory management, CRM, sales, accounting/finance, import/logistics, and approvals. Then integrate AI Car Match, vehicle comparison, loan estimates, brokerage, customs services, and sell-or-swap workflows.

This document is an initial requirements baseline. Before implementation, validate business policies, legal and import requirements, payment processes, customer terms, and accounting treatment with the company's relevant stakeholders and professional advisers.